

# SendIt API — End-to-End Test & Verification Suite

Interactive Swagger UI Testing Walkthrough, Verification Signals, and RBAC Matrix

## 1. Test Account Credentials & RBAC Matrix

| Role    | Username      | Password        | Access Rights                                   |
|---------|---------------|-----------------|-------------------------------------------------|
| Admin   | admin_user    | AdminPass123!   | Full System Control, Webhook Reg, All Documents |
| Manager | manager_nyeri | ManagerPass123! | Manual Weather Enrichment, Search All Docs      |
| Staff   | staff_rider   | StaffPass123!   | Upload Documents, Search Own Documents          |

## 2. Detailed Interactive Test Execution Suite

### Test 1: User Authentication & JWT Authorization

|                   |                                                                                                                                                                                                      |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope & Purpose:  | Verifies bcrypt password checking, token creation, and Swagger UI Bearer header injection.                                                                                                           |
| Swagger UI Steps: | 1. Open <a href="http://localhost:8000/docs">http://localhost:8000/docs</a> 2. Click green 'Authorize' button3. Enter Username: staff_rider, Password: StaffPass123!4. Click 'Authorize' and 'Close' |
| Expected Status:  | 200 OK                                                                                                                                                                                               |
| Sample Payload:   | { "access_token": "eyJhbGciOiJIUzI1Ni...", "token_type": "bearer"}                                                                                                                                   |

### Test 2: Document Upload & Automatic Weather Enrichment

|                   |                                                                                                                                                                                 |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope & Purpose:  | Verifies rate limiting (10/hr), file disk storage, geocoding, and Open-Meteo weather API fetch.                                                                                 |
| Swagger UI Steps: | 1. Expand POST /documents/upload2. Attach waybill file (.pdf, .png, or .docx)3. Enter city: Nyeri, country: Kenya4. Description: 'Waybill for coffee transport'5. Click Execute |
| Expected Status:  | 201 Created                                                                                                                                                                     |
| Sample Payload:   | { "message": "Upload complete", "document_id": 1, "status": "enriched"}                                                                                                         |

### Test 3: Exercise 1 — Multi-Criteria Document Search

|                   |                                                                                                                                                            |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope & Purpose:  | Verifies query filtering (q, city, status, date range) and user data isolation for Staff role.                                                             |
| Swagger UI Steps: | 1. Expand GET /documents/search2. Set q: 'coffee' and city: 'Nyeri'3. Click Execute                                                                        |
| Expected Status:  | 200 OK                                                                                                                                                     |
| Sample Payload:   | [ { "id": 1, "original_filename": "waybill.pdf", "city": "Nyeri", "status": "enriched", "weather_data": "{\"temperature\": 21.5, \"windspeed\": 12.3}" } ] |

#### Test 4: Exercise 2 — Document Version Control

|                   |                                                                                                                           |
|-------------------|---------------------------------------------------------------------------------------------------------------------------|
| Scope & Purpose:  | Verifies document revision incrementing and automatic toggling of is_latest status flag.                                  |
| Swagger UI Steps: | 1. Expand POST /documents/upload/versioned2. Select file with SAME filename as Test 23. Set city: 'Nyeri'4. Click Execute |
| Expected Status:  | 200 OK                                                                                                                    |
| Sample Payload:   | { "message": "Version uploaded", "version": 2, "document_id": 2}                                                          |

#### Test 5: Exercise 3 & RBAC — Webhook Registration Security

|                   |                                                                                                                                                                             |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope & Purpose:  | Verifies Role-Based Access Control blocking non-admin users from registering webhooks.                                                                                      |
| Swagger UI Steps: | 1. Expand POST /webhooks/register2. Set webhook_url: 'https://webhook.site/test'3. Set event_type: 'document.enriched'4. Click Execute (while authenticated as staff_rider) |
| Expected Status:  | 403 Forbidden                                                                                                                                                               |
| Sample Payload:   | { "detail": "Admin authorization required"}                                                                                                                                 |